

Problem Set 12
Comp 301
Spring 2025

Week 12: 23.05.2025 - 24.05.2025

Submit your answers in a pdf file. Name it as ID_username.pdf with your ID and username
(Example: 1234567_mbabelli22.pdf). You are expected to submit your work by May 24, 2025 midnight. **Read the questions carefully. Good luck!**

Problem 1: What does the following program return if we run the program using

- (a) Call by Value?
- (b) Call by Reference?

```
let reset = proc (x) set x = 0 in
  let inc = proc (x) set x = -(x, -1) in
    let a = 0 in
      begin
        (inc a);
        (inc a);
        (reset a);
        (inc a);
        a
      end
    end
```

Problem 2: What does the following program return if we run the program using

- (a) Call by Value?
- (b) Call by Reference?

```
let add = proc (x) set x = -(x, -2) in
  let sub = proc (x) set x = -(x, 1) in
    let y = 4 in
      begin
        (add y);
        (sub y);
        (add y);
        y
      end
    end
```

Problem 3: The following program is executed using two different evaluation strategies:

- Mode A: Call by Value
- Mode B: Call by Reference

What is the final value of a under each mode?

```
let foo = proc (x) set x = -(x, -2) in
  let bar = proc (x) set x = -(x, -( -(0, x), x)) in
    let a = 1 in
      let b = a in
        begin
          (foo a);
          let a = 5 in
            (bar b);
            (foo b);
            a
          end
        end
      end
    end
  end
```

- (a) What does the program return under call by value?
- (b) What does the program return under call by reference?

Question 1

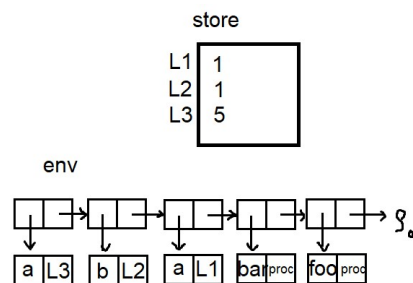
- (a) 0, we can see we pass a to the functions but in the end return just a, and since this is call by value, the value referenced by a is not changed, it is copied when we call the procs.
- (b) 1, first 2 incs set the value referenced by a to 1, then 2, then reset sets it to 0, then inc sets it to 1.

Question 2

- (a) 4, same reason as q1
- (b) 7, first add sets the value referenced by y to 6, the next sub sets it to 5, the last add sets it to 7.

Question 3

- (a) 1, foo and bar do not change the value referenced by a, while the innermost let overshadows it in the environment without deleting the previous value or reference, so when the scope of innermost let ends, a still references the value 1.
- (b) 3, since let creates a new value at a new reference
 (foo a) -> a=3
 let a = 5; -> a=5, b=1
 (bar b) -> b=3
 (foo b) -> b=5
 a -> 3



env and store before (foo b) is evaluated